

Answers (teacher copy)

Level 1 · Set A (mean)

1. 4
2. 7
3. 4
4. 20
5. 5
6. 6
7. 5
8. 15
9. 6
10. 10

Level 1 · Set B (median, mode)

1. 3, 3
2. 4, 4
3. 5, 2
4. 6, 6
5. 11, 10
6. 7, 9
7. 14, 13
8. 3, 3
9. 22, 25
10. 7, 7

Level 2 · Set C

1. 5.5
2. 14.5
3. 5.5
4. 4
5. 2 and 6
6. 151 cm
7. mean 22, median 22, modes 21 and 24
8. mean 2, median 2, mode 2
9. 10.4
10. mean \$20, median \$20, mode \$20

Level 3 · Set D

1. $24 - 19 = 5$
2. $360 - 293 = 67$
3. e.g. 4, 4, 5, 8, 9 (many answers possible)
4. mean 12, median 6. The median is better because 40 is an outlier that pulls the mean up.
5. mean goes from 7 to about 9.17 (up by about 2.17); median goes from 7 to 7.5
6. $40 - 30 = 10$
7. $30 - 5 - 9 = 16$
8. $90 - 70 = 20$
9. Mode (size 8). It shows which size sells most often.
10. Both have a mean of 16. Class B has a higher median (16 vs 15).

Extension

1. Total 650 → 665, new mean 66.5
2. 2, 7, 8, 9, 9 · 3, 6, 8, 9, 9 · 4, 5, 8, 9, 9 (the two smallest must be different, less than 8 and add to 9)
3. (a) 16, 17, 18 (b) $n - 1$, n , $n + 1$ add to $3n$, and $3n \div 3 = n$, the middle number

Teacher examples

1. Ex 1: 6
2. Ex 2: median 5, mode 4
3. Ex 3: 7.5
4. Ex 4: mean 13, median 12.5, no mode
5. Ex 5: $40 - 30 = 10$
6. Ex 6: mean ≈ 17.3 , median 12. The median is better (44 is an outlier).

Exit ticket

1. Level 1: 8
2. Level 2: median 6.5, mode 3
3. Level 3: $36 - 24 = 12$